

THE HODGE CONJECTURE FOR FERMAT FOURFOLDS OF ODD DEGREE AT MOST 199

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ABSTRACT. Let $X_m^4 = \{x_0^m + \cdots + x_5^m = 0\} \subset \mathbb{P}^5$ be the Fermat fourfold of degree m . We prove the Hodge conjecture for X_m^4 for every odd $m \leq 199$.

The proof introduces three closure mechanisms for Hodge $(2, 2)$ -characters. First, if the character multiset splits into two zero-sum triples, then the corresponding rational Hodge block is transported from a $(1, 1)$ -substructure of a product of Fermat curves and is therefore algebraic. Second, algebraicity follows for characters that, after adjoining two vanishing pairs, decompose into an Aoki standard sextuple and a grade-2 Hodge quadruple. Third, the exceptional class at $m = 33$ has a lift to level 66 that is quasi-decomposable; its algebraicity then descends along the finite morphism $X_{66}^4 \rightarrow X_{33}^4$.

Together with the known decomposable, quasi-decomposable, and standard cases, these criteria cover all odd degrees $m \leq 199$. This is established by a complete computer-assisted census, accompanied by a completeness proof for the enumeration, re-verifiable per-orbit certificates (positive witnesses and, for the beyond-machinery orbits, the negative screenings), and an algorithmically independent reimplementing reproducing the census in full. Finally, an exact Jacobi-sum computation at $p = 67$ shows that no cycle defined over $\mathbb{Q}(\zeta_{33})$ projects nontrivially onto the exceptional $m = 33$ block. More generally, for every finite extension of $\mathbb{Q}(\zeta_{33})$ over which a certifying cycle is defined, every residue degree above 67 is divisible by 6.

1. INTRODUCTION AND MAIN RESULTS

Let $X_m^4 \subset \mathbb{P}^5$ be the Fermat fourfold $x_0^m + \cdots + x_5^m = 0$, $G = \mu_m^6/\mu_m$, and for a character $a = (a_0, \dots, a_5) \in (\mathbb{Z}/m)^6$ with $\sum a_i \equiv 0$, $a_i \neq 0$, let $V(a) \subset H_{\text{prim}}^4$ be the (1-dimensional) eigenspace; a is a Hodge $(2, 2)$ character iff $\sum_i \langle ta_i \rangle = 3m$ for all $t \in (\mathbb{Z}/m)^\times$, where $\langle x \rangle \in \{1, \dots, m-1\}$ is the residue (Shioda [9]). The known algebraicity machinery is: (i) the *induced structure* (Shioda [9], Shioda–Katsura [10]), which transports algebraicity through decompositions—da Silva’s conditions $(P1)/(P2)$ are quasi-decomposability in the semigroup M_m [3]; and (ii) *standard cycles* (Aoki [1]). da Silva [3, Prop. 3.4 (= Prop. 3.6 of the arXiv version)] found the first Hodge class beyond both at $m = 33$ and proposed a candidate cycle $[W]$ (Question 1). (His closures — $m \leq 20$ [Thm. 2.7], m prime or $m = 4$ [Thm. 2.6], $m = p^2$ [Thm. 2.8], $m = 21, 27$ [Thm. 3.3], and fourfolds of degree $m \leq 100$ coprime to 6 [Prop. 3.1], numbering of the published version [3] — leave the composite non-square degrees with $3 \mid m$, beyond his 21, 27, untouched; every class closed below lies there.) In summary:

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mechanism	source	covers
induced structure	Shioda [9]; Shioda–Katsura [10]	decomposable, quasi-decomposable
standard cycles; lattice calculus	Aoki [1, 2]	standard characters; $\nu \in S_m$; Thm. 0.1 degree forms
computational closures	da Silva [3]	$m \leq 20$; m prime or 4; p^2 ; 21, 27; $m \leq 100$ coprime to 6
Theorems 1.1–1.3 + census	this paper	the residual odd orbits through 199

We work in his semigroup formalism and prove:

Notation. Throughout $m \geq 3$ ($m \leq 2$ gives a hyperplane or a smooth quadric, where the conjecture is classical). A multiset a of nonzero residues with $\sum a_i \equiv 0 \pmod{m}$ has *grade* g if $\sum_i \langle ta_i \rangle = gm$ for *every* $t \in (\mathbb{Z}/m)^\times$ (the constancy is the Hodge condition: grade-3 sextuples are the Hodge (2, 2) characters of X_m^4 , grade-2 quadruples the (1, 1) characters of the surface X_m^2); $\sim$ is equality of multisets up to coordinate permutation, and the *content* of a class $a = (a_0, \dots, a_5)$ is $\gcd(a_0, \dots, a_5, m)$ (content- e classes are inflations from level m/e). $\text{claim}_m(a)$ (or $\text{claim}(a)$ when m is clear) is Aoki’s CLAIM(a): $V(a)$ is spanned by algebraic cycle classes [1, Introduction] — equivalently, some *rational* algebraic cycle class has nonzero $V(a)$ -component, $\omega_a(Z) \neq 0$. Three levels are kept distinct throughout: eigenlines are indexed by *ordered* tuples ($V(b) \neq V(b')$ for distinct ordered characters even when they agree as multisets; coordinate permutations are realized by automorphisms of X_m^4 , which is why claim is invariant under them, and it is likewise a Galois-orbit-level property); $M(a) := \bigoplus_b V(b)$ is the rational Galois block — the sum over the distinct *ordered* conjugates $b = ta$, $t \in (\mathbb{Z}/m)^\times$, their number being the index of the ordered stabilizer $\{t : ta = a \text{ coordinatewise}\}$, *not* the number of multiset classes; and the census classifies sorted representatives modulo both actions — legitimate since claim is invariant under both, but never identified with an eigenline count. $\omega_a(Z)$ denotes the $V(a)$ -component of the cycle class of Z (Aoki’s averaged projector [1, Introduction]). $*$ is Aoki’s juxtaposition — concatenation of the entry tuples; characters of X_m^r and X_m^s concatenate to one of X_m^{r+s+2} ($n = r + s + 2$ in Aoki’s indexing by cohomological degree), grades adding [1, §1] — and, for even s , $\mathfrak{D}_m^s$ his set of $(s + 2)$ -entry characters that are coordinate permutations of $(a_0, -a_0, \dots, a_{s/2}, -a_{s/2})$ ($s/2 + 1$ vanishing pairs), represented by linear cycles [1, §1, Thm. 1-1]; only $\mathfrak{D}_m^2$ is used below. M_m is da Silva’s additive semigroup of solutions $(x_1, \dots, x_{m-1}; y)$, $y > 0$, of $\sum_i \langle ti \rangle x_i = my$ for all t (multiplicities intrinsic), $M_m(g)$ its grade- g part; a class is *decomposable* if it splits in M_m and *quasi-decomposable* if it splits after adjoining one element of $M_m(1)$ — da Silva’s (P1)/(P2) [3, Def. 2.4]. $\#$ is his pairing: for $\beta = (b_0, \dots, b_{r+1})$, $\gamma = (c_0, \dots, c_{s+1})$ with nonzero entries and $b_{r+1} + c_{s+1} \equiv 0$ (the set $U_m^{r,s}$), $\beta \# \gamma = (b_0, \dots, b_r, c_0, \dots, c_s)$ [3, before Cor. 2.3]. $\sigma_{p,x}$ denotes the standard characters (Proposition 1.6). $u(a, p)$ is the scalar by which geometric Frobenius at a split prime p acts on $V(a)(n/2)$ — the Tate-normalized Jacobi sum $j(a)/p^{n/2}$ (the Jacobi-sum count of diagonal hypersurfaces, [11, pp. 500–502]; [12]) — so a divisorial class has $u = 1$ at split primes whose Frobenius fixes the class of a certifying divisor (the second $m = 39$ class of §3 calibrates this in-text). Three symbols are kept apart: $u(a, p)$ above is the *local* normalized scalar; χ_a is the associated *global* Jacobi-sum Hecke character *in its Tate-normalized form*: Weil’s unnormalized character J_a [12] has $|J_a(\mathfrak{p})| = N\mathfrak{p}^{n/2}$ in every embedding (n the cohomological degree of the ambient Fermat variety), so J_a itself is *not* of finite order, and $\chi_a := J_a/N^{n/2}$ is its finite-order part — every use of χ_a in this paper is of the normalized character, whose value at a split prime is the local scalar $u(a, p)$; it is multiplicative under juxtaposition, $\chi_{\alpha*\beta} = \chi_\alpha\chi_\beta$ (the normalizing powers add along the weights; the machine-verified instances are the displayed u -identities); and $\nu(a) \in \mathbb{Z}^{m-1}$ is Aoki’s *multiplicity vector*,

additive under juxtaposition, $\nu(\alpha * \beta) = \nu(\alpha) + \nu(\beta)$ (Aoki writes u for ν). S_m denotes Aoki's lattice: the subgroup of $\mathbb{Z}^{m-1}$ generated by the $\nu(\delta)$ over the vanishing pairs $\delta = \{k, m-k\}$ and the $\nu(\sigma_{p,x})$ over Aoki's standard elements for all primes $p \mid m$ — for odd p the AP family of Proposition 1.6, at even m also Aoki's $p = 2$ family $\sigma_{2,x} = (x, x + \frac{m}{2}, m - 2x, \frac{m}{2})$ [2, §2] — the “standard calculus”; this generating family is verbatim the generator set of Aoki's lattice S_m (Proposition 7.1), with membership decided exactly by integer linear algebra (Hermite normal form), and B_m^4 the grade-3 Hodge-character set of the Fermat fourfold at level m ; in lattice statements we freely identify a with $\nu(a)$ ($a \in B_m^4 \setminus S_m$ means $\nu(a) \notin S_m$ — a statement about the formal calculus, not about the character group). The witness $w = (1, 4, 16, 22, 25, 31)$ below is, up to coordinate permutation, the Galois conjugate $7w$ of da Silva's displayed representative $(7, 10, 13, 19, 22, 28) \sim 7w$ of [3, Prop. 3.4 (= Prop. 3.6 of the arXiv version)].

Theorem 1.1 (*-split closure). *Let a be a Hodge $(2, 2)$ character of X_m^4 , m arbitrary, whose 6-multiset splits as two zero-sum triples $\beta' \uplus \gamma'$. Then the rational Hodge classes of $M(a)$ are algebraic: each is the image, under the algebraic joining-line correspondence of the Shioda–Katsura structure map, of a $\mathbb{Q}$ -linear combination of divisor classes on $X_m^1 \times X_m^1$ furnished by Lefschetz $(1, 1)$ (no single ruled-surface representative is claimed).*

In particular the classes at $m = 39, 117, 195$ (two orbits each), none quasi-decomposable or standard, are closed; the orbit of $(1, 7, 16, 22, 34, 37)$ is a certified gap family ($\nu \notin S_m$ at 39 and at its inflations 117, 195 — all verifier-backed), not covered by the published mechanisms compared in §9 (to the author's knowledge), while the second orbit's ν lies in S_{39} (an explicit six-generator ± 1 certificate), so its algebraicity is also derivable from the standard calculus (Lemma 4.1(iii) of [2]) — the closure certificates and the join-transport construction scheme are new for both.

Theorem 1.2 (two-pair standard closure). *Let a be a Hodge $(2, 2)$ character of X_m^4 admitting vanishing pairs $p_1 = \{k, m-k\}$, $p_2 = \{\ell, m-\ell\}$ with*

$$a \uplus p_1 \uplus p_2 = S \uplus Q \quad (\text{equality of 10-element multisets}),$$

where S is an Aoki 5-standard sextuple and Q a grade-2 Hodge quadruple. Then the rational Hodge classes of $M(a)$ are algebraic.

This applies to—and, by exhaustive search, uniquely presents—the three remaining $5 \mid m$ classes of the census:

m	a	p_1, p_2	$S \uplus Q$
45	$(1, 19, 20, 28, 30, 37)$	$(5, 40), (10, 35)$	$(1, 10, 19, 28, 37, 40) \uplus (5, 20, 30, 35)$
105	$(3, 24, 50, 66, 85, 87)$	$(15, 90), (45, 60)$	$(3, 24, 45, 66, 87, 90) \uplus (15, 50, 60, 85)$
105	$(1, 22, 43, 64, 90, 95)$	$(5, 100), (20, 85)$	$(1, 22, 43, 64, 85, 100) \uplus (5, 20, 90, 95)$

—none quasi-decomposable, standard, or Theorem 1.1-splittable: closing, with the induced $m = 135$ copy, the $p = 5$ -standard-block classes of the census. Of these, the $\mathbb{Q}(\zeta_{21})$ orbit at 105 is a certified gap class, not covered by the published mechanisms compared in §9 (to the author's knowledge); $m = 45$ and 135 fall under the degree forms of [2, Thm. 0.1(i)], and the $\mathbb{Q}(\zeta_7)$ orbit's ν lies in S_{105} , so for those three the algebraicity is also derivable from [2] and the explicit two-pair presentations are the new content.

Theorem 1.3 (level-lifted closure; the $m = 33$ witness). *Let $w = (1, 4, 16, 22, 25, 31)$, the unique non-quasi-decomposable non-standard Hodge class of X_{33}^4 (uniqueness is the census's, used for Theorems 1.4 and 1.5; the proof below uses only the displayed identities), and $a = 2w \bmod 66$ its induced copy on X_{66}^4 . Then*

$$a \uplus \{1, 65\} \uplus \{25, 41\} = Q \uplus S, \quad Q = (1, 25, 44, 62), \quad S = (2, 8, 32, 41, 50, 65),$$

with Q a grade-2 Hodge quadruple and S quasi-decomposable at level 66 via $S \uplus \{33, 33\} = (2, 32, 33, 65) \uplus (8, 33, 41, 50)$, the self-paired element $(33, 33) \in M_{66}(1)$ being its unique quasi-witness. Hence $V(a)$ is algebraic (Lefschetz + da Silva + Aoki 1-4, as in Theorem 1.2), and $V(w)$ is algebraic by descent along $\pi: X_{66}^4 \rightarrow X_{33}^4$, $(x_i) \mapsto (x_i^2)$ (Lemma 5.2).

Theorem 1.4 (completion of the odd-degree census; $m \leq 199$). *For every odd $m \leq 199$, every grade-3 Hodge $(2, 2)$ class of X_m^4 is algebraic: the classes beyond decomposable $\cup$ quasi $\cup$ standard are exhausted by Theorems 1.1, 1.2, 1.3 and level transport (the inflation/descent pair). In particular the Hodge conjecture holds in full for every Fermat fourfold of odd degree ≤ 199 : in codimension 2 the non-primitive summand of H^4 is spanned by the algebraic class h^2 , so the statement follows from the primitive one; codimension 1 is Lefschetz $(1, 1)$; and codimension 3 follows from codimension 1 by hard Lefschetz — $L^2: H^2(\mathbb{Q}) \xrightarrow{\sim} H^6(\mathbb{Q})$ is an isomorphism of Hodge structures, so every $(3, 3)$ Hodge class is L^2 of a $(1, 1)$ Hodge class, and L^2 of a divisor class is algebraic. (Codimensions 0 and 4 are trivial: H^0 and H^8 are spanned by the classes of X and of a point.)*

Theorem 1.5 (field obstruction; Question 1). *Let $a_0 = (7, 10, 13, 19, 22, 28) = 7w$ be da Silva's representative of the $m = 33$ orbit. No algebraic cycle defined over $\mathbb{Q}(\zeta_{33})$ has nonzero projection onto the Galois orbit of $V(a_0)$: Frobenius at any prime above 67 acts on every conjugate eigenline by a primitive sixth root of unity, of exact order 6 — an exact certificate $u(a_0, 67) = 1 + \zeta_3 = \zeta_6$ (for the pinned character choice of §8) computed in $\mathbb{Z}[x]/\Phi_{33}$ with no floating point. (Under an alternative Jacobi-sum definition $j_{\text{alt}} = -j$ the formula relating j_{alt} to geometric Frobenius acquires the compensating sign; the geometric eigenvalue remains $\zeta_6^{\pm 1}$.) Since W has $\mathbb{Q}$ -coefficients, $[W]$ and all its translates have zero projection: the answer to Question 1 is NO at $m = 33$.*

Consequently the obstruction is local and exact: if a cycle defined over a finite extension $L/\mathbb{Q}(\zeta_{33})$ — not assumed Galois over $\mathbb{Q}(\zeta_{33})$; the constraint is per prime — has nonzero projection, then at every prime $\mathfrak{q} \mid \mathfrak{p} \mid 67$ of L one needs $u(a_0, 67)^{f(\mathfrak{q}/\mathfrak{p})} = 1$ — no hypothesis on the ramification of $L/\mathbb{Q}(\zeta_{33})$ is needed: the eigenline's character is unramified at 67 over the base, so inertia at $\mathfrak{q}$ acts trivially and Frobenius acts through $u^{f(\mathfrak{q}/\mathfrak{p})}$ — so $6 \mid f(\mathfrak{q}/\mathfrak{p})$: every residue degree of L above 67 is divisible by 6. (The same machinery, run at the closed levels, computes values of order 9 at $m = 45$ and of orders 42 and 7 at $m = 105$ at sampled split primes — computational observations, not part of any theorem here; they indicate that the cycles of Theorem 1.2 likewise cannot be $\mathbb{Q}(\zeta_m)$ -rational.)

Proposition 1.6 (standard = AP). *Let m be odd. Aoki's standard characters $\sigma_{p,x}$ for an odd prime $p \mid m$ — defined for x with $d/\gcd(x, d) > 2$, $d = m/p$ [1, §1] — are exactly the padded arithmetic-progression characters $\{x, x + m/p, \dots, x + (p-1)m/p, -px\}$ with $px \not\equiv 0 \pmod{m}$, of grade $(p+1)/2$; grade-3 standard classes exist iff $5 \mid m$ and $m > 5$.*

Verification status. Theorem 1.1: proved (transport pinned to the Shioda–Katsura primary text). Theorem 1.2: proved (a citation chain through Aoki's Theorems 1-1, 1-4(i),(ii), 2-1 [1] + Lefschetz $(1, 1)$ + an inflation lemma; all identities verified exactly, including $u(a, p) = u(S, p)u(Q, p)$). Theorem 1.4: machine-verified census, reproduced exactly by an algorithmically independent implementation (the four-tier certificate structure is laid out in §6). Theorem 1.5: proved, exact certificate. Proposition 1.6: read off the primary construction. We give no explicit defining equations for a cycle on X_{33}^4 (Theorem 1.3's certifying cycles are constructed at level 66 and transported by the finite morphism).

2. STANDARD CYCLES ARE THE AP FAMILY (PROPOSITION 1.6)

For odd $p \mid m$, $d = m/p$, and x with $d/\gcd(x, d) > 2$, the standard character is the multiset $\{x, x+d, \dots, x+(p-1)d, m-px\}$ of grade $(p+1)/2$ (Aoki [1, §1]; da Silva [3], Appendix). For odd m the admissibility hypothesis is equivalent to the visible one: d is odd, so $d/\gcd(x, d) \leq 2$ forces $d \mid x$, i.e. $px \equiv 0 \pmod{m}$ — exactly the degenerate case excluded in the statement. Grade 3 forces $p = 5$; at $m = 5$ every candidate degenerates ($5x \equiv 0$), whence “ $m > 5$ ”. The identification is load-bearing for Theorem 1.4: it is what makes the census’s standard classifier complete for odd m (Lemma 6.1). The padded arithmetic-progression family is exactly the witness family of the field-of-definition threshold of the companion papers: at the surface level in [4] (with explicit Hasse–Davenport witnesses), and in general even dimension in *Galois-invariant ranks of middle algebraic cycles on Fermat varieties* (in preparation). $\square$

Remark 2.1 (even m). For even m (used only in the even census of §7) the admissibility hypothesis is strictly stronger than $px \not\equiv 0$: it excludes $x \equiv d/2 \pmod{d}$. Every excluded multiset contains a vanishing pair — $(x + k_1d) + (x + k_2d) \equiv 0 \pmod{m}$ reduces to $1 + 2j + k_1 + k_2 \equiv 0 \pmod{5}$ for $x = d/2 + jd$, solvable with $0 \leq k_1 < k_2 \leq 4$ since $k_1 + k_2$ ranges over all residues mod 5 — so it is decomposable, hence algebraic for the trivial reason, and the census classifier (which tests decomposability first) is unaffected. (Only $p = 5$ yields grade-3 sextuples, so this is the only case the classifier meets.)

3. THE *-SPLIT CLOSURE THEOREM (THEOREM 1.1)

The proof factors through the following transport lemma, which records separately the passage from complex eigenlines to rational algebraic classes: which objects are correspondences, over which fields they are defined, where the Tate twist sits, and why the scalar onto $V(a)$ is nonzero.

Lemma 3.1 (rational transport). *Let $C = X_m^1$ (m arbitrary), $a = \beta' * \gamma'$ a Hodge $(2, 2)$ character split into two zero-sum triples, and let f denote the second summand of the Shioda–Katsura structure map [3, Thm. 2.2] (going back to Shioda’s inductive structure [9]; the geometric join construction is [10, Thm. 1.7]), a G -equivariant morphism of Hodge structures*

$$f: H_{\text{prim}}^1(C, \mathbb{Q}) \otimes H_{\text{prim}}^1(C, \mathbb{Q}) \longrightarrow H_{\text{prim}}^4(X_m^4, \mathbb{Q})$$

of type $(1, 1)$ — equivalently, a weight-preserving $(0, 0)$ -morphism out of $H^1 \otimes H^1(-1)$; this (-1) is the only Tate twist in the construction. Then:

- (i) *at every $t \in (\mathbb{Z}/m)^\times$ the line $V(t\beta') \otimes V(t\gamma')$ has Hodge type $(1, 0) \otimes (0, 1)$ or $(0, 1) \otimes (1, 0)$, so the block $B(a) := \bigoplus V(t\beta') \otimes V(t\gamma') \subset H^2(C \times C, \mathbb{C})$ — the sum over the distinct ordered paired characters $(t\beta', t\gamma')$, pairs of ordered triples, whose coordinatewise stabilizer may be nontrivial (e.g. $\beta' = (3, 3, 3)$ at $m = 9$) and coincides with the ordered stabilizer of $a = \beta' * \gamma'$, so the summands of $B(a)$ and $M(a)$ biject — is purely of type $(1, 1)$;*
- (ii) *$B(a)$ is Galois-stable, so $B(a)_{\mathbb{Q}} := B(a) \cap H^2(C \times C, \mathbb{Q})$ satisfies $B(a)_{\mathbb{Q}} \otimes \mathbb{C} = B(a)$, and by Lefschetz $(1, 1)$ every element of $B(a)_{\mathbb{Q}}$ is a $\mathbb{Q}$ -combination of divisor classes on the surface $C \times C$;*
- (iii) *f is induced by an algebraic correspondence (the blow-up–quotient–blow-down chain of [10], defined over the cyclotomic field of the join construction), and it is a G -equivariant isomorphism onto its structural summand [3, Thm. 2.2]: it maps each one-dimensional eigenline $V(t\beta') \otimes V(t\gamma')$ isomorphically — in particular by a nonzero scalar — onto $V(ta)$, hence carries $B(a)$ onto the corresponding block of $M(a)$. On decomposable algebraic inputs $f(Z_1 \otimes Z_2) = m Z_1 \wedge Z_2$, the ruled join by lines [3, Thm. 2.2(c)]; this*

formula is used only for the geometric description of the image cycles, not for the scalar on the H^1 -eigenlines, which are not spanned by algebraic classes;

- (iv) consequently every rational Hodge class of $M(a)$ is the class of an algebraic cycle: it is the f -image of a $\mathbb{Q}$ -combination of divisor classes — the image of an algebraic class under the algebraic joining-line correspondence, hence an algebraic cycle class over $\mathbb{Q}$, lying in $H^4(X_m^4, \mathbb{Q})$ by construction. Field-of-definition bookkeeping: the correspondence above is defined over a cyclotomic field, and no $\mathbb{Q}$ -rationality of the cycles is claimed or needed — algebraicity of a rational Hodge class is a $\overline{\mathbb{Q}}$ -statement (the fields the certifying cycles live over are the subject of Theorem 1.5, not of this lemma).

Proof. (i) is the grade bookkeeping $|t\beta'| + |t\gamma'| = 3$, each summand in $\{1, 2\}$ (verified exactly for every printed instance in the ancillary identities). (ii): Galois permutes the summands of $B(a)$ (the index set is one orbit), so $B(a)$ is defined over $\mathbb{Q}$ as a sub-Hodge structure; being purely $(1, 1)$, its rational part is divisorial by Lefschetz $(1, 1)$. (iii) is [3, Thm. 2.2] (cf. [9], [10, Thm. 1.7]): f is a G -equivariant isomorphism onto its image, so its restriction to each one-dimensional eigenline is injective, i.e. a nonzero scalar, and equivariance matches the source and target characters. (iv) combines (i)–(iii). (An explicit block projector — an orbit sum of graph correspondences with rational, Galois-symmetric coefficients, algebraic over $\mathbb{Q}$ — can be written down, but is not needed: the class handed to f already lies in $B(a)_{\mathbb{Q}}$.) $\square$

Proof of Theorem 1.1. A split $a = \beta' \uplus \gamma'$ into zero-sum triples realizes $a \sim \beta' * \gamma'$, and by multiplicity one $M(a)$ is the f -image of the block $B(a)$ of Lemma 3.1. By the lemma, every rational Hodge class of $M(a)$ is the f -image of a $\mathbb{Q}$ -combination of divisor classes on $C \times C$, with a nonzero scalar on each eigenline (part (iii) of the lemma) — algebraic, because the image of an algebraic class under an algebraic correspondence is algebraic. Geometrically the correspondence is the joining-line construction — over $(z, w) \in C \times C$ the line $\lambda z + \mu w$ lies on X_m^4 : $\sum(\lambda z_i)^m + \sum(\mu w_j)^m = \lambda^m \cdot 0 + \mu^m \cdot 0$ — and on decomposable algebraic inputs its effect is the ruled join $f(Z_1 \otimes Z_2) = m Z_1 \wedge Z_2$ [3, Thm. 2.2(c)]; for a general divisor no single ruled-surface representative of the image class is claimed, and only the correspondence-image statement is used. $\square$

Remark 3.2. Quasi-decomposability tests only decompositions with both factors Hodge; curves carry no Hodge classes in H^1 , so the $(P1)/(P2)$ formalism is structurally blind to this odd $\otimes$ odd branch. At $m = 39$: $(1, 7, 16, 22, 34, 37) = (1, 16, 22) \uplus (7, 34, 37)$ and $(1, 14, 16, 22, 29, 35) = (1, 16, 22) \uplus (14, 29, 35)$; the second has $j(a) = p^2$ exactly at every computed split prime — consistent with a divisorial certificate whose class is fixed by the Frobenius elements in question (algebraicity alone does not force this: a divisor defined only over an extension can carry a nontrivial finite character), the calibration promised in the Notation.

4. THE TWO-PAIR CLOSURE THEOREM (THEOREM 1.2)

Proof. Write $\delta = p_1 * p_2 = (k, m-k, \ell, m-\ell) \in \mathfrak{D}_m^2$ (Aoki's all-pairs classes, represented by linear cycles with nonzero self-pairing, [1, Thm. 1-1]). The multiset identity gives $a * \delta \sim S * Q$ on X_m^8 ($\sim$ a coordinate permutation, an algebraic automorphism). (1) Q is a grade-2 Hodge quadruple: its rational Galois block on the Fermat surface is purely of type $(1, 1)$, hence divisorial by Lefschetz $(1, 1)$, and $\text{claim}(Q)$ follows. (2) For the $\sigma_{5,1}$ blocks ($m = 45, d = 9$ and $m = 105, d = 21$; $\gcd(1, d) = 1$) $\text{claim}(S)$ is [1, Thm. 2-1]; for the $\sigma_{5,3}$ block at $m = 105$ note $S = 3 \cdot \sigma_{5,1}$ at level 35 and apply Lemma 4.1. (In general a 5-standard block $\sigma_{5,x}$ with $g = \gcd(x, d) > 1$ is the g -inflation of $\sigma_{5,x/g}$ at level m/g , where the gcd is 1, so Lemma 4.1 and Thm. 2-1 cover every admissible block.) (3) $\text{claim}(S * Q)$ is [1, Thm. 1-4(i)] ($r = 4, s = 2, n = 8$). (4) $\text{claim}(a)$ is

[1, Thm. 1-4(ii)]: if there exists $\delta \in \mathfrak{D}_m^s$ with $\text{claim}(\alpha * \delta)$, then $\text{claim}(\alpha)$ —applied with $\alpha = a$, $s = 2$. $\square$

Lemma 4.1 (inflation). *Let $m = e m'$ and β a level- m' character with every entry of $e\beta$ nonzero mod m . Then $\text{claim}_{m'}(\beta) \Rightarrow \text{claim}_m(e\beta)$.*

Proof. $\pi: X_m^n \rightarrow X_{m'}^n$, $(x_i) \mapsto (x_i^e)$ is a morphism with $\pi(\zeta x) = \zeta^e \pi(x)$ for $\zeta \in \mu_m$, so π^* maps $V(\beta)$ into $V(e\beta)$ equivariantly (the nonzero-entries hypothesis is what makes $e\beta$ a legitimate primitive character, so the target eigenline exists); π finite surjective gives $\pi_* \pi^* = (\deg \pi) \text{id} \neq 0$ on $\mathbb{Q}$ -cohomology, so $\pi^* \xi$ spans $V(e\beta)$, and for a representing cycle Z , $\omega_{e\beta}(\pi^* Z) = \pi^*(\omega_\beta(Z)) \neq 0$. (Here $\pi^* Z$ is the pullback in $CH_{\mathbb{Q}}$: π is a finite surjective morphism of smooth projective varieties, hence flat by miracle flatness, and flat pullback is compatible with cycle classes.) $\square$

Remark 4.2. da Silva’s quasi-decomposability adjoins one pair and splits $4+4$; the two-pair $6+4$ pattern is outside the definition. The three presentations were found by exhaustive iterated-closure search and are unique per class; equivalently the three classes are sum-preserving two-entry surgeries of standard sextuples $((10, 40) \mapsto (20, 30)$ at 45; $(45, 90) \mapsto (50, 85)$, $(85, 100) \mapsto (90, 95)$ at 105). The characters multiply exactly: $\chi_a = \chi_S \chi_Q$, i.e. $u(a, p) = u(S, p) u(Q, p)$ at every split prime. The surviving $m = 33$ class resists this genre exhaustively (no augmentation by up to five vanishing pairs admits a certifying partition; $5 \nmid 33$, so no standard blocks exist).

5. LEVEL-LIFTED CLOSURE OF THE WITNESS (THEOREM 1.3)

Proof. All identities are exact (machine-verified; the ancillary re-derives each). The augmented multiset partitions as stated; Q is grade-2 Hodge. For S : the pair $(33, 33)$ is a legitimate element of $M_{66}(1)$ ($\langle 33t \rangle + \langle 33t \rangle = 66$ for every odd t ; multiplicities are permitted), and in da Silva’s coordinates $S = \beta \# \gamma$ with $\beta = (2, 32, 65; 33)$, $\gamma = (8, 41, 50; 33)$ — an element of his $U_{66}^{2,2}$, the distinguished entries 33, 33 summing to 0 mod 66 (the definition preceding [3, Cor. 2.3]); Corollary 2.3(b) with $(r, s) = (2, 2)$ and Lefschetz $(1, 1)$ for β, γ give $\text{claim}(S)$. The hypotheses of Corollary 2.3(b) are exactly: r, s even, algebraicity of both factors, nonzero entries, and the matching condition — each verified above; the self-paired value $b_3 = c_3 = 33 = m/2$ is admissible (Remark 5.1). Then, exactly as in Theorem 1.2: $\text{claim}(S * Q)$ by [1, Thm. 1-4(i)]; $S * Q \sim a * \delta$ with $\delta = (1, 65, 25, 41) \in \mathfrak{D}_{66}^2$; $\text{claim}(a)$ by [1, Thm. 1-4(ii)]. Descent is Lemma 5.2. $\square$

Remark 5.1 (self-pair admissibility). The hypotheses of the pairing statement used above — membership in $U_{66}^{2,2}$ and Corollary 2.3(b) of [3] — are: even r, s , nonzero entries, algebraicity of both factors, and the matching condition $b_3 + c_3 \equiv 0$. Pairwise distinctness of the entries is not among them, and the underlying structure isomorphism [3, Thm. 2.2] carries no nondegeneracy hypothesis on the characters; the self-paired distinguished value $b_3 = c_3 = 33 = m/2$ is therefore legitimate. This is the one point where the even level is essential: no self-pair exists at an odd level (Remark 5.3(iv)).

Lemma 5.2 (descent). *Let $m = e m'$ and β a level- m' character with every entry of $e\beta$ nonzero mod m . Then $\text{claim}_m(e\beta) \Rightarrow \text{claim}_{m'}(\beta)$.*

Proof. $\pi: X_m^n \rightarrow X_{m'}^n$, $(x_i) \mapsto (x_i^e)$ is a finite surjective morphism ($\sum x_i^m = \sum (x_i^e)^{m'}$ identically) intertwining the group actions along the e -th-power surjection $G_m \twoheadrightarrow G_{m'}$; hence $\pi_* V(e\beta) \subseteq V(\beta)$ (again the nonzero-entries hypothesis makes $e\beta$ a legitimate primitive character). Since $\pi_* \pi^* = (\deg \pi) \text{id} \neq 0$ and $\pi^* V(\beta) = V(e\beta)$ with both eigenlines 1-dimensional, $\pi_*: V(e\beta) \xrightarrow{\sim} V(\beta)$. Proper pushforward carries algebraic cycle classes to algebraic cycle classes ($\text{cl} \circ \pi_* = \pi_* \circ \text{cl}$), and π is finite of relative dimension zero, so there is no degree shift or Tate twist. $\square$

Remark 5.3. (i) With the inflation lemma (Lemma 4.1), claim is level-transparent: $\text{claim}_{m'}(\beta) \Leftrightarrow \text{claim}_m(e\beta)$ — every lift of an open class is an independent chance to close it. (ii) The mechanism is selective: the two-pair-at-the-double search does not fire for the induced copies of the $m = 54$ classes at 108 or of the $m = 50$ class at 100 (all of these close at their own levels at depth 3–4; even-census receipts, §7), and the beyond-machinery $m = 66$ classes resist the single two-pair at level 66 and its lift 132 (they too close at depth 3; even-census receipts, §7). (iii) Consistency with the field obstruction (§8): the constructed cycles (Lefschetz divisors on X_{66}^2 , Aoki’s linear spaces with $\varepsilon = \zeta_{132}$, radicals) are not $\mathbb{Q}(\zeta_{66})$ -rational, and $\mathbb{Q}(\zeta_{66}) = \mathbb{Q}(\zeta_{33})$ — the obstruction forbids exactly the $\mathbb{Q}(\zeta_{33})$ -rational certificates, and this theorem produces non-rational ones (their precise minimal field of definition is not computed here). The local Frobenius values agree exactly: $u(2w@66, 67) = \zeta_6 = u(w@33, 67)$. (iv) The uniqueness of the quasi-witness $v = 33$ shows the self-paired $m/2$ element — which exists only at even levels — is strictly load-bearing: this is why the exhaustive level-33 search (five augmentation pairs, all partition types) could not see the closure.

6. THE CENSUS AND ITS COMPLETENESS (THEOREM 1.4)

For each odd m : enumerate grade-3 Hodge multisets (meet-in-the-middle on half-unit sum profiles); reduce to Galois orbits; classify decomposable $\rightarrow$ quasi $\rightarrow$ standard (Prop. 1.6) $\rightarrow$ *-split (Thm. 1.1) $\rightarrow$ survivors; the surviving orbits are then closed by Theorems 1.2/1.3 and level transport. The engine reproduces the Shioda/da Silva anchors ((P_m) at $m = 9, 21, 27$; the $m = 33$ witness) and was reproduced, on all 89 odd levels of the census list ($21 \leq m \leq 199$, $m \neq 23$; the omitted small levels and $m = 23$ are classical [3, Thms. 2.6–2.7]), by a second, algorithmically independent implementation (structured-array meet-in-the-middle, cross-validated against its own brute force for $m \leq 143$). After Theorem 1.1 (which closes the *-split classes at 39, 117, 195), seven beyond-machinery orbits remain: four primitive ($m = 33, 45, 105 \times 2$) and the induced copies at 99, 135, 165; after Theorem 1.2 (which closes 45 and both 105, hence by Lemma 4.1 also $m = 135$) and Theorem 1.3 (which closes 33, hence 99 and 165), nothing remains. Induced copies are tied to their primitives by the finite pull–push correspondences of Lemmas 4.1 and 5.2, which transfer claim in both directions.

Lemma 6.1 (census completeness). *Fix m and consider the shipped generator, classifiers, and canonizer.*

- (i) *Every element of M_m has even length $2g$, g its grade: for nonzero entries $\langle(m-t)x\rangle = m - \langle tx\rangle$, so $\sum_i \langle(m-t)a_i\rangle = Lm - \sum_i \langle ta_i\rangle$ and grade-constancy forces $g = L/2$. In particular $M_m(1)$ consists exactly of the vanishing pairs $\{k, m-k\}$ (including the self-pair $k = m/2$ at even m), and a grade-3 sextuple is decomposable exactly when it contains a vanishing pair — the complementary quadruple is then automatically grade-2 Hodge.*
- (ii) *The quasi-decomposability test is complete for da Silva’s definition: for a pair-free sextuple a , the 8-multiset $a \uplus \{k, m-k\}$ splits in M_m only with part-lengths $(2, 6)$ or $(4, 4)$, by (i); in every $(2, 6)$ split the pair is either $\{k, m-k\}$ itself or $\{a_i, m-a_i\}$ with $k \in \{a_i, m-a_i\}$, and in both cases the split is the trivial one (the pair part is the adjoined pair, the sextuple part is a itself) — excluded by the non-triviality clause of [3, Def. 2.4]. Hence a is quasi-decomposable iff some $a \uplus \{k, m-k\}$ splits into two grade-2 Hodge quadruples — exactly the test performed.*
- (iii) *The meet-in-the-middle generator produces exactly the grade-3 Hodge sextuples: the join condition enforces $\sum_i \langle ta_i\rangle = 3m$ for every t in a half-unit system, and the complement identity of (i) extends it to all units; conversely every sorted sextuple is the join of its two sorted halves, each of which occurs in the enumeration of all 3-multisets of nonzero residues.*

- (iv) *Canonization retains the lexicographically least sorted conjugate — a system of distinct representatives for the $(\mathbb{Z}/m)^\times$ -orbits.*

*Consequently the census verdicts (decomposable / quasi / standard / *-split / survivor) coincide with their definitions level by level; the standard classifier is complete for odd m by Proposition 1.6.*

Proof. (i) is the two displayed identities; for the last claim, a vanishing pair contributes m to every unit sum, so the complementary quadruple has constant grade 2. (ii): the parts have even lengths summing to 8 with grades (1, 3) or (2, 2); in a (2, 6) split the length-2 part is a vanishing pair inside $a \uplus \{k, m - k\}$, and since a is pair-free it is $\{k, m - k\}$ (complement = a) or uses one adjoined entry, forcing $k \in \{a_i, m - a_i\}$ and complement $(a \setminus \{a_i\}) \uplus \{a_i\} = a$ (multiplicities included; the even- m self-pair case is identical). (iii): if the joined profile sums to $3m$ on the half-units t , then at $m - t$ it sums to $6m - 3m = 3m$; the converse containment is the split of a sorted sextuple at position 3. (iv): distinct orbits have distinct minima. The implementations are additionally calibrated against direct brute force (the independent reimplementations for $m \leq 143$; the even-parity engine, exact match of representative counts, for $m \leq 14$) and reproduce the Shioda/da Silva anchors above. The full 89-level run — per-level counts, classification tallies, survivor lists, and per-orbit closure witnesses — ships as checksummed data artifacts with the ancillary files. $\square$

What is certified, and how. Four tiers, in decreasing logical strength. (1) *Per-orbit certificates*: for each of the 78,299 orbit representatives across the 89 levels, a stored witness of its classification (the vanishing pair / the quasi split / the standard parameter / the *-split triples), re-verified against the definitions on every run of the supplied runner; for the seven beyond-machinery orbits the runner re-establishes the *negative* screenings live (no vanishing pair, no quasi witness at any k , not standard, no zero-sum split), so the terminal labels are re-checked certificates, not labels. (2) The *completeness lemma* above, which reduces “no orbit is missed” to the definitional correctness of the shipped generator and canonizer. (3) The *independent reimplementations*: an algorithmically independent second program regenerates the census from scratch and matches it level by level (its complete 89-level log ships; the runner re-executes it live at anchor levels and regenerates a fresh sub-range against the summary table below). (4) *Checksummed receipts* of the historical campaign runs: a checksum certifies file integrity, never recomputation — recomputation is what tiers (1)–(3) execute. A per-level summary table (`census_level_summaries.json`: representative count, tally by kind, survivor list, and the SHA-256 hash of the canonically sorted representative list) makes “no missed orbit” checkable level by level: any regeneration, by any implementation, must reproduce the count and the hash.

The bound 199. The bound is a computational horizon, not a structural one: every closure mechanism used here is degree-uniform, and nothing in the method stops at 199. What grows is the census cost — on the reference machine (Apple M4 Max, Python 3.13) the engine takes 158 s at the single level $m = 199$, the full two-engine 89-level sweep of the pinned receipt took ≈ 36 minutes, and the complete independent rerun ≈ 12 minutes — and 199 was chosen to include, with margin, the largest levels at which new primitive phenomena appear in the odd census: the two-pair genre at 105 (the last new primitive beyond-machinery orbit) and the *-split gap family through $195 = 5 \cdot 39$. The probes beyond the bound (next paragraph) found no new phenomenon.

Additional single-engine censuses at the induced-risk levels beyond 199 — computational observations: these three levels sit outside the witness-tier receipts of Theorem 1.4 — found: at $231 = 33 \cdot 7$ and $297 = 33 \cdot 9$ one beyond-machinery orbit each, the induced copy of the $m = 33$ class (content 7, resp. 9), closed by Theorem 1.3 and transport; at the control level $273 = 3 \cdot 7 \cdot 13$, two *-split classes (closed by Theorem 1.1) and nothing else. They indicate that the conclusion

of Theorem 1.4 also holds at $m = 231, 273, 297$; the theorem itself does not depend on these levels.

7. COMPUTATIONAL OUTLOOK: THE EVEN-DEGREE BOUNDARY THROUGH DEGREE 250

The odd-degree theorems are complete. The even-degree boundary is a different landscape, and this closing section only *reports its computed map* — engines, closure ledger, certificates, and pinned receipts ship in the ancillary — deferring systematic treatment (full statements and proofs of the closure and wall structure) to the companion paper [5], with which this note shares its ancillary package. Nothing in Theorems 1.1–1.5 or in the odd census depends on this section, and every computational claim below is backed receipt-by-receipt within the ancillary of *this* submission — the companion adds statements and proofs, not data. All censuses use the parity-corrected decomposability test (the self-paired element $m/2$; part (i) of Lemma 6.1 covers both parities); every lattice verdict below carries an independent certificate in the ancillary — a modular kernel witness for each non-membership, a re-summed integer combination for each membership. One statement is load-bearing also for the odd-degree novelty classification above, and is therefore proved here.

Proposition 7.1 (the implemented lattice is Aoki’s S_m). *Aoki [2, §2] defines A_m as the zero-sum sublattice of the free abelian group on $\mathbb{Z}/m \setminus \{0\}$; u — our ν — as the multiplicity-vector map, additive under juxtaposition, $\nu(\alpha * \beta) = \nu(\alpha) + \nu(\beta)$; the standard elements, verbatim, as $\sigma_{p,a} = (a, a + \frac{m}{p}, \dots, a + \frac{(p-1)m}{p}, m - pa)$ for $p \geq 3$ and $\sigma_{2,a} = (a, a + \frac{m}{2}, m - 2a, \frac{m}{2})$ for $p = 2$, over primes $p \mid m$ ($m > p$) and $0 < a < \frac{m}{p}$ with nonzero entries; $\mathfrak{S}_m$ as the set of α admitting $\alpha * \delta \sim \sigma_1 * \dots * \sigma_k * \delta'$ with δ, δ' unions of vanishing pairs, with $\mathfrak{D}_m \subset \mathfrak{S}_m$; and $S_m \subset A_m$ as the subgroup generated by $\nu(\mathfrak{S}_m)$. Consequently S_m equals the $\mathbb{Z}$ -span of the pair vectors $\nu(\{k, m - k\})$ and the standard vectors $\nu(\sigma_{p,a})$, $p = 2$ included — exactly the generator list of the implemented lattice — so the machine verdicts $\nu \in S_m / \nu \notin S_m$ are statements about Aoki’s group itself.*

Proof. $\supseteq$: every pair class lies in $\mathfrak{D}_m \subset \mathfrak{S}_m$, and every standard element σ satisfies $\sigma * \delta \sim \sigma * \delta$, so $\sigma \in \mathfrak{S}_m$: each listed generator is in $\nu(\mathfrak{S}_m)$. $\subseteq$: if $\alpha * \delta \sim \sigma_1 * \dots * \sigma_k * \delta'$, then $\nu(\alpha) = \sum_i \nu(\sigma_i) + \nu(\delta') - \nu(\delta)$, an integer combination of the listed generators. The two spans coincide. $\square$

The computed map (computed; engines, closure ledger, and receipts in the ancillary; full statements and proofs in [5]). For even $m \leq 108$ the census finds twenty-one primitive beyond-machinery orbits (beginning at $m = 50$; the odd law “survivor $\Rightarrow 3 \mid m$ ” fails in the even sector); through degree 200 it finds fourteen more, and through degree 250 exactly three more — thirty-eight base-tier primitives in all. Iterated partition certificates of the Theorem 1.2/1.3 kind close all but ten; an exhaustive exchange scan — claim-equivalence moves $T = S \uplus A \leftrightarrow T' = S \uplus B$, $|A| = |B| \in \{2, 3\}$, with exchanger $q = A \uplus (-B)$ a grade-2 Hodge quadruple (Lefschetz, unconditional) or a grade-3 sextuple of known claim, *every* move out of *every* open class enumerated at its own level and its lifts — organizes the residual into claim-equivalence walls, and two further classes then close. The $m = 168$ class falls under Aoki’s published Theorem 0.1(i) [2] ($168 = 2^3 \cdot 3 \cdot 7$ has his degree form; his §4 proof opens with the bridge quoted in the Appendix). Citing it, we record two errata of the printed text, both verified against the primary and neither affecting the theorem as stated: the companion CMUSP 51 (2002) paper misassigns the statement, and the $m = 28$ gap-generator entry on p. 185 is a misprint (not a zero-sum character) — a machine-verified *possible replacement* of exactly the required join form is $\xi_{28} = (1, 9, 18) * (10, 21, 25)$, with $\nu \notin S_{28}$, $2\nu \in S_{28}$ (whether it is the intended generator modulo S_{28} is not established; the 168 closure does not depend on it). The isolated degree-210

class $(1, 79, 109, 121, 151, 169)$ closes by an explicit 40-generator ± 1 lattice certificate $\nu \in S_{210}$ — pairs and standard elements only, converted into a literal Aoki chain by the negation closure — whence V is algebraic by Theorems 1-4(i),(ii) of [1]; the full signed combination ships in the ancillary and is displayed, block by block, in [5]. What remains *unresolved by the implemented closure family and the cited published theorems* through even degree 250 is eight certified gap classes — all with $\nu \notin S_m$, $2\nu \in S_m$ (the doubling holds for every Hodge class by [2, Thm. 2.1]), all outside the degree forms of Aoki’s theorem — organized in three claim-equivalence walls: W_{70} (the champion $(1, 20, 24, 42, 61, 62)$ at 70 and its exchange partner at 210), W_{110} (three members), W_{114} (three members). Every lattice verdict carries an independent certificate in the ancillary (a modular kernel witness for each “no”, the re-summed ± 1 combination for the “yes”); within the scanned move family the walls are exchange-isolated, and closing any member closes its wall. Gap-ness bounds the *standard* calculus, not partition search: the $\mathbb{Q}(\zeta_{21})$ class at 105 closes by Theorem 1.2 while remaining a gap class. The wall table, the per-class dossiers, the deep negative sweeps, and the independent even-parity verification are in [5].

8. THE FIELD OBSTRUCTION AND QUESTION 1 (THEOREM 1.5)

Normalization. Fix a split prime $p \equiv 1 \pmod m$ and a multiplicative character $\chi: \mathbb{F}_p^\times \rightarrow \overline{\mathbb{Q}}^\times$ of exact order m (extended by $\chi(0) = 0$). Set

$$S_0(a) = \sum_{\substack{v \in (\mathbb{F}_p^\times)^6 \\ v_0 + \dots + v_5 = 0}} \prod_i \chi^{a_i}(v_i), \quad j(a) = \frac{S_0(a)}{p-1} \in \mathbb{Z}[\zeta_m]$$

(the scaling $v \mapsto \lambda v$ is free since $\sum a_i \equiv 0$, so S_0 is exactly divisible by $p-1$; the certificate asserts this). By Weil’s count of points of diagonal hypersurfaces [11, pp. 500–502], [12], *geometric* Frobenius at p acts on the étale eigenspace $V(a) \subset H_{\text{prim}}^4$ by $j(a)$, with $|j(a)| = p^2$ in every embedding — in our even dimension ($n = 4$) the point-count sign is $+1$ — hence on $V(a)(2)$ by $u(a, p) = j(a)/p^2$ (arithmetic Frobenius acts by the inverse scalar; every statement in this paper is about geometric Frobenius); the conjugate eigenlines carry $u(ta, p)$, computed by replacing χ with χ^t . Under an inhomogeneous convention ($\sum v_i = 1$) the Jacobi symbol changes by a sign that the point-count formula compensates; the certificate’s asserted $(u')^3 = -1$ pins the geometric order regardless. A 30-digit numeric cross-check against the Gauss-sum product $\prod_i g(\chi^{a_i})/p^3$ corroborates the normalization (ancillary; corroboration only — the certificate itself is exact). The certificate’s character is pinned: χ is the character of $\mathbb{F}_{67}^\times$ with $\chi(g) = \zeta_{33}$ on the smallest primitive root $g = 2$ (the shipped code computes discrete logarithms base 2); replacing χ by a conjugate character permutes the twenty conjugate values and can exchange $\zeta_6 \leftrightarrow \zeta_6^{-1}$ at $t = 1$ — the convention-independent assertions are the exact order six and the two-value distribution.

Proof. Fix a prime $\ell \neq 67$ and an embedding $\iota: \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$; cohomology in this proof is ℓ -adic étale cohomology with $\overline{\mathbb{Q}}_\ell$ -coefficients (the eigenspaces correspond under comparison), and the algebraic integers $u(ta_0, 67) \in \mathbb{Z}[\zeta_{33}]$ are read in $\overline{\mathbb{Q}}_\ell$ through ι ; the asserted identities $(u')^3 = -1$, $(u')^6 = 1$, $u' \neq 1$ hold in $\mathbb{Z}[\zeta_{33}]$, hence under every embedding, so no choice matters. The ℓ -adic class of a cycle over K is G_K -fixed. Take $\mathfrak{p} \mid 67$ in $\mathbb{Q}(\zeta_{33})$ ($67 \equiv 1 \pmod{33}$ splits, good reduction). $\mu_{33} \subset \mathbb{F}_{67}$ makes the G -action $\mathbb{F}_{67}$ -rational, so Frobenius preserves each $V(ta_0)$ and acts on $V(ta_0)(2)$ by $u(ta_0, 67)$ as fixed in the Normalization above. An exact integer dynamic program in $\mathbb{Z}[x]/\Phi_{33}$ gives $u(a_0, 67) = 1 + \zeta_3 = \zeta_6$, and the 20 unit-indexed values $u(ta_0, 67)$, $t \in (\mathbb{Z}/33)^\times$ — the Frobenius scalars on the *twenty* conjugate eigenlines of $M(a_0)$: the ordered stabilizer is trivial ($7t \equiv 7$ forces $t = 1$), the multiset stabilizer $\langle 4 \rangle$ has order 5, so the twenty eigenlines fall into four coordinate-permutation classes of five; the scalar — a symmetric function

of the entries — is constant on each class, and the four class-values coincide in pairs — take exactly the two values $\zeta_6^{\pm 1}$, each on ten eigenlines. The certificate asserts *exact order six* on every conjugate — $(u')^3 = -1$, $(u')^6 = 1$, $u' \neq 1$ — and the geometric scalar is independent of the Jacobi-sum sign convention, so each eigenvalue is a primitive sixth root of unity, hence $\neq 1$. A Frobenius-fixed vector scaled by $\neq 1$ vanishes componentwise. $\square$

W has $\mathbb{Q}$ -coefficients and its translates are $\mathbb{Q}(\zeta_{33})$ -rational; by the census the $m = 33$ class is the unique non-quasi non-standard class at that degree, so its projection vanishes and Question 1 is answered negatively. The YES-branch would have proved the Hodge conjecture at these classes; the NO says only that this candidate cannot certify it, and imposes a sharp local residue-degree constraint on any number field of definition of a certifying cycle.

9. THE CLOSED CLASSES' DOSSIERS

The $m = 33$ witness (closed by Theorem 1.3): indicated eigenvalue field $\mathbb{Q}(\sqrt{-3})$ (a computed label), all entries $\equiv 1 \pmod 3$; its *Tate-normalized* Jacobi-sum Hecke character χ_a is of finite order — for a grade-constant (Hodge) character the infinity type of Weil's unnormalized J_a is the norm power N^2 itself, so $\chi_a = J_a/N^2$ has trivial infinity type (cf. [12]) — and its computed values at split primes lie in μ_6 : kernel primes occur ($p = 859$: every conjugate value exactly 1), and order 6 is attained ($p = 67$, the prime Theorem 1.5 uses; only the certified per-prime obstruction is used, and no global order is claimed). The dossier's remaining role is the acceptance test for explicit cycles (Theorem 1.3's pass by construction). A Weil-class alternative would need a precise carrier: Weil classes on abelian *fourfolds* of Weil type are now unconditionally algebraic (Markman [6], with Schoen's degeneration argument [8]), but the Shioda–Katsura realization carries $V(w)$ on CM abelian varieties of dimension ≥ 10 (forced by the faithful $\mathbb{Q}(\zeta_{33})$ -action), and no algebraic compression onto a $\mathbb{Q}(\sqrt{-3})$ -Weil fourfold is known — the route stays conditional on that open compression, and is now unnecessary.

The closed classes $m = 45, 105$: their exact eigenvalue fields $\mathbb{Q}(\zeta_9), \mathbb{Q}(\zeta_7), \mathbb{Q}(\zeta_{21})$ (degrees 6, 6, 12) exceed the imaginary-quadratic multiplication field required by the published algebraicity theorems for Weil-type classes known to us (Schoen [8]; Markman [6]); the CM-field secant-sheaf program [7], which would reach larger fields, is explicitly conditional on a semiregularity “not addressed yet” (loc. cit.); on the canonical CM carriers (dimensions 12 and 24) restriction of scalars to an imaginary quadratic subfield shifts the cohomological degree away from 4, so no descent to the covered case exists. For the $\mathbb{Q}(\zeta_{21})$ class at 105, Theorem 1.2's Fermat-internal route is, among the mechanisms compared here, at present the only proof of algebraicity known to the author (a gap class: $\nu \notin S_{105}$, $2\nu \in S_{105}$); the $m = 45$ and $\mathbb{Q}(\zeta_7)$ $m = 105$ classes are also reachable by the standard calculus — $\nu \in S_{45}$, resp. $\nu \in S_{105}$, by explicit four-generator ± 1 certificates (re-derivable via the ancillary lattice test), hence algebraic by Lemma 4.1(iii) of [2] as well, Theorem 1.2 supplying the explicit two-pair presentation. The computed per-prime values (orders 9/42/7 at sampled split primes; computational observations) indicate analogous local residue-degree constraints at the closed levels; explicit equations remain open.

10. METHODS, CODE AVAILABILITY, AND PROVENANCE

Exact objects throughout (Gauss/Jacobi sums as algebraic integers; the decisive certificate in $\mathbb{Z}[x]/\Phi_{33}$ with no floating point; the exact-twist certificate framework follows the surface companion [4]); numerics corroborate or measure invariant lower bounds only. Every proof-critical computational claim of Theorem 1.4 carries either a stored per-orbit witness certificate or an algorithmically independent reimplementations; the runner re-executes the anchor levels live and re-verifies every stored witness and every supplied certificate against the definitions. The census engines, the closure oracles, the lattice membership test, the exact Frobenius certificate,

and a standalone verifier of every closure identity in this paper are provided as ancillary files (`anc/`; `run_all.sh` reproduces the smoke tier — identities, census anchors, both lattice verdicts, the Theorem 1.5 certificate, the engine calibrations, the re-verification of all stored per-orbit witnesses including the survivor negatives, the independent census at anchor levels, and the level-summary cross-check on a fresh sub-range — in under a minute on the reference machine whose measured runtimes the ancillary README reports; `run_full_census.sh` is the archival tier, the complete from-scratch independent regeneration of all 89 levels against the pinned summary hashes; the README records the reference configuration and the measured runtimes quoted in §6), together with the deep-closure ledger and the stored degree-210 certificate as data artifacts; an algorithmically independent census reimplement is bundled — its complete 89-level run ships as a checksummed receipt, and the runner re-executes it live at the anchor levels — and the original implementation used during verification is methodologically independent of both. This manuscript was prepared with AI assistance; all computational claims are backed by the ancillary code and receipts, and the author takes full responsibility for all proofs, code, and bibliographic claims.

APPENDIX A. THE CITED AOKI STATEMENTS

For self-containedness we reproduce the exact statements used, in the numbering of the primary texts.

From [1] (notation as there: $\mathfrak{B}_m^n$ the Hodge characters of X_m^n , $\mathfrak{D}_m^n$ the classes that are coordinate permutations of $(a_0, -a_0, \dots, a_r, -a_r)$, $r = n/2$; $*$ juxtaposition; CLAIM(α): $V(\alpha)$ is generated by classes of algebraic cycles on X_m^n):

- **Theorem 1-1** (Shioda). *The linear space L represents $\delta \in \mathfrak{D}_m^n$; more precisely $\omega_\delta(L) \cdot \overline{\omega_\delta(L)} = (-1)^r m^{n+1}$.*
- **Theorem 1-4** (Shioda, Ran). *Let r and s be non-negative even integers such that $n = r + s + 2$, and let $\alpha \in \mathfrak{B}_m^r$, $\beta \in \mathfrak{B}_m^s$. Then: (i) if CLAIM(α) and CLAIM(β) are true, then CLAIM($\alpha * \beta$) is also true; (ii) if there exists $\delta \in \mathfrak{D}_m^s$ such that CLAIM($\alpha * \delta$) is true, then CLAIM(α) is also true.*
- **Theorem 2-1**. *For an odd prime $p \mid m$, $d = m/p$, $\gcd(a, d) = 1$, the variety Y defined by (2.1) there is a subvariety of X_m^{p-1} of codimension $r = (p-1)/2$ representing $\alpha = \sigma_{p,a}$; more precisely $\omega_\alpha(Y) \cdot \overline{\omega_\alpha(Y)} = (-1)^r p^{p-2} m^p$.*
- **§1 definition**. *For a prime $p \mid m$, $d = m/p$, and each i with $d/(i, d) > 2$: $\sigma_{p,i} = (i, i + d, \dots, i + (p-1)d, m - pi)$ for $p \geq 3$ — the (p -)standard elements.*

From [2] (verified verbatim against the open-access copy in the Rikkyo University repository, doi:10.14992/00009819; §7 records two errata of the printed text, both likewise verified against the primary — the ξ_{28} list entry on p. 185, and the attribution line after Theorem 1.2 of the companion CMUSP 51 (2002) paper, which assigns its part (ii) (= Theorem 0.1(i) below) to the Catalan-curves reference):

- **Theorem 0.1**. *Suppose the prime factorization of m is one of the following forms: (i) $m = 2^a 3^b 5^c 7^d$, where a, b, c, d are non-negative integers such that either $c = 0$ or $d = 0$; (ii) $m = p^e$ or $2p^e$, p an odd prime. Then the Hodge conjecture is true for all abelian varieties of Fermat type of degree m . Its proof (§4 there) opens: “In view of Corollary 3.2, it suffices to show that the Hodge conjecture for X_m^n is true for all n ” — the bridge to the Fermat varieties themselves used in §7; the verification then reduces, via Lemma 4.1 and Theorem 2.1 there, to the gap generators ξ_m at $m = 12, 15, 20, 21, 28$ (the $m = 28$ entry being the misprint recorded in §7).*
- **Lemma 4.1** (verbatim). *Let $\alpha, \beta \in B_m$. Then: (i) if both $V(\alpha)$ and $V(\beta)$ are algebraic, then so is $V(\alpha * \beta)$; (ii) if $V(\alpha * \delta)$ is algebraic for some $\delta \in D_m$, then so is $V(\alpha)$;*

(iii) if $\alpha \in S_m$, then $V(\alpha)$ is algebraic. No hypothesis on m is imposed; the groups S_m and the “gap group” are Aoki’s own objects (§2 there), and Proposition 7.1 identifies the implemented lattice with S_m generator-by-generator (his §2 also proves, via Theorem 2.1, that $2\alpha \in S_m$ for every $\alpha \in B_m$ — the doubling our $2u$ -certificates instantiate).

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